

TEST CASE CATALOGUE

Structured validation coverage for the VirtuBox task application.

Test Case ID	Module	Test Scenario	Precondition	Test Steps	Test Data	Expected Result	Type (Positive/Negative/Edge)	
TC-REG-01	Registration	Register with valid details	Email is not registered	Submit the registration form with valid values	Name, unique email, valid password	Account is created once	Positive	
TC-REG-02	Registration	Register with existing email	Account already exists	Submit the same registration again	Existing email	Request is rejected and no duplicate account is created	Negative	
TC-REG-03	Registration	Invalid email format	Registration page is available	Submit the form with malformed email	user@, user.example.com	Validation error is returned; account is not created	Negative	
TC-REG-04	Registration	Missing or empty required fields	Registration page is available	Submit with blank fields, one at a time and all together	Empty name, email, or password	Request is rejected with field-level errors	Negative	
TC-REG-05	Registration	Weak password or boundary length	Password policy exists or is defined	Try values at, below, and above the allowed limit	Short password and maximum-length password	Values follow the documented password rule	Edge	
TC-REG-06	Registration	Registration input with spaces/special characters	Registration page is available	Submit a name with spaces and supported punctuation	Anne-Marie O'Neil	Valid supported data is stored safely and consistently	Edge	
TC-LOG-01	Login	Login with valid credentials	Registered user exists	Submit the correct email and password	Registered email/password	Login succeeds and a session/token is issued	Positive	
TC-LOG-02	Login	Incorrect password	Registered user exists	Submit the wrong password	Correct email, wrong password	Login fails with a safe error	Negative	
TC-LOG-03	Login	Unknown user	No account uses the email	Submit login details	missing@example.com	Login fails without exposing unnecessary account details	Negative	
TC-LOG-04	Login	Missing or invalid fields	Login page is available	Submit blank email, blank password, and malformed email	Empty values; bad-email	Validation or authentication error is returned; no session is created	Negative	
TC-LOG-05	Login	Protected task list while logged out	No valid session exists	Request the task list directly	No token	Request is rejected or redirected to login	Negative	
TC-LOG-06	Login	Logout ends access	User is logged in	Logout, then request the task list	Valid session before logout	Old session/token cannot access protected tasks	Positive	
TC-CRT-01	Create task	Create a valid task	User is logged in	Submit a task and then load the list	Buy milk	Task is created and appears once	Positive	
TC-CRT-02	Create task	Missing, empty, or whitespace-only task	User is logged in	Submit without a title or with spaces only	"" , " "	Request is rejected; no blank record is saved	Negative	
TC-CRT-03	Create task	Long or unexpected task input	User is logged in	Submit text at and over the configured limit	Long title, special characters	Limit and encoding rules are applied; server does not crash	Edge	
TC-CRT-04	Create task	Create task without authentication	No valid session exists	Send a create request directly	Valid task payload, no token	Request is rejected and no task is created	Negative	
TC-READ-01	View tasks	View existing tasks	Logged-in user has tasks	Request the task list	Two saved tasks	User sees the correct tasks and values	Positive	
TC-READ-02	View tasks	View an empty list	Logged-in user has no tasks	Request the task list	No saved tasks	Empty list or clear empty-state message is returned	Positive	
TC-READ-03	View tasks	User cannot view another user's tasks	Two users have separate tasks	Log in as User B and request the list	User A and User B task data	Only User B's tasks are returned	Negative	
TC-UPD-01	Edit task	Edit an existing task	User owns a task and is logged in	Change the title and save; reload the list	Old title to New title	New value is saved and displayed	Positive	
TC-UPD-02	Edit task	Edit with missing/empty title	User owns a task	Submit an update with blank title	""	Update is rejected and old value remains	Negative	
TC-UPD-03	Edit task	Update a missing task	User is logged in	Submit update for an unknown ID	does-not-exist	Not-found response is returned; no other task changes	Negative	
TC-UPD-04	Edit task	Update another user's task	User A owns the task; User B is logged in	Submit update using User A's ID	User A task ID	Request is rejected and User A's task is unchanged	Negative	
TC-DEL-01	Delete task	Delete an existing task	User owns a task and is logged in	Delete the task; reload the list	Existing task ID	Delete succeeds and task no longer appears	Positive	
TC-DEL-02	Delete task	Delete a missing task	User is logged in	Delete an unknown ID	does-not-exist	Not-found response is returned without deleting another task	Negative	
TC-DEL-03	Delete task	Delete without authentication	No valid session exists	Send a delete request directly	Existing task ID, no token	Request is rejected and task remains	Negative	
TC-ERR-01	Error handling	Malformed request	API is available	Send invalid JSON, wrong field type, or unknown ID	title: [], broken JSON	Controlled 4xx response is returned; server does not crash	Negative	
TC-ERR-02	Error handling	API/database failure	Failure can be simulated in a test environment	Load or save while service returns 500/timeout	Simulated failure	Clear error is returned and data is not falsely reported as saved	Negative	

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TC-ERR-03	Error handling	Duplicate request	User is logged in	Repeat a create or delete request quickly	Same payload/ID twice	No unintended duplicate or unrelated deletion occurs	Edge	